Typically, the AS will be running on your own internal network and issued with a certificate by a private CA for your organization, so it’s unnecessary to trust all of these public CAs.

- The default TLS settings include a wide variety of *cipher suites* and protocol versions for maximum compatibility. Older versions of TLS, and some cipher suites, have known security weaknesses that should be avoided where possible. You should disable these less secure options and re-enable them only if you must talk to an old server that cannot be upgraded.

TLS cipher suites

A TLS ***cipher suite*** is a collection of cryptographic algorithms that work together to create the secure channel between a client and a server. When a TLS connection is first established, the client and server perform a *handshake*, in which the server authenticates to the client, the client optionally authenticates to the server, and they agree upon a session key to use for subsequent messages. The cipher suite specifies the algorithms to be used for authentication, key exchange, and the block cipher and mode of operation to use for encrypting messages. The cipher suite to use is negotiated as the first part of the handshake.

For example, the TLS 1.2 cipher suite `TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256` specifies that the two parties will use the Elliptic Curve Diffie-Hellman (ECDH) key agreement algorithm (using ephemeral keys, indicated by the final E), with RSA signatures for authentication, and the agreed session key will be used to encrypt messages using AES in Galois Counter Mode. (SHA-256 is used as part of the key agreement.)

In TLS 1.3, cipher suites only specify the block cipher and hash function used, such as `TLS_AES_128_GCM_SHA256`. The key exchange and authentication algorithms are negotiated separately.

The latest and most secure version of TLS is 1.3, which was released in August 2018. This replaced TLS 1.2, released exactly a decade earlier. While TLS 1.3 is a significant improvement over earlier versions of the protocol, it’s not